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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	SPNG
Project / Solution Name	e-Salary Guard
Name (Leader)	Dixson Oswaldo Tanjaya
Email & Phone (Leader)	dixsonoswaldooo@gmail.com | +60 1115442413
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://sdc-2026-spng.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/dixsonoswaldooo-oss/sdc2026-spng

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

e-Salary Guard is a single Next.js application deployed on Replit. Users sign in through the browser to one of two role separated areas, employee or HR; the session is a signed httpOnly cookie. Server side route handlers check the signed in role, the employee's ownership and the input on every change, then read and write a managed PostgreSQL database with node postgres. The demo state employees, synthetic attendance (standing in for Orbyte Attendance and eLeave), overtime and leave requests, pay rules, locked payroll snapshots, messages and the activity log is stored as JSONB in one table, with a version number so simultaneous changes cannot overwrite each other.

After every change the server re-runs the payroll engine, a unit tested TypeScript module, and records new deduction events. Each event produces a simulated WhatsApp message and a log entry.

For leave requests, the route /ai/classify-leave sends only the reason text to SEA-LION (Gemma SEA-LION-v3-9B-IT) on RunPod Serverless and checks the reply; if the call fails or is slow, a keyword fallback is used. HR reviews every label before it affects pay. Keys are Replit Secrets. The diagram is Appendix A.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Next.js 16.4	Full stack framework: pages, routing, server route handlers	MIT
React 19.3	User interface components	MIT
TypeScript 5.9	Typed language for the app and payroll engine	Apache 2.0
Tailwind CSS 4.3	Styling and responsive layout	MIT
node postgres (pg) 8.23	Database queries from the server	MIT
Drizzle ORM 0.45 / Drizzle Kit 0.31	Declares the database table and creates it on Replit	Apache 2.0

Name & Version	Purpose in Project	License
Vitest 5.0	73 unit tests: payroll engine, server permissions, translations	MIT
Inter (Google Fonts)	Typeface	SIL OFL 1.1

3. APIs & Cloud Services

3.1 External APIs

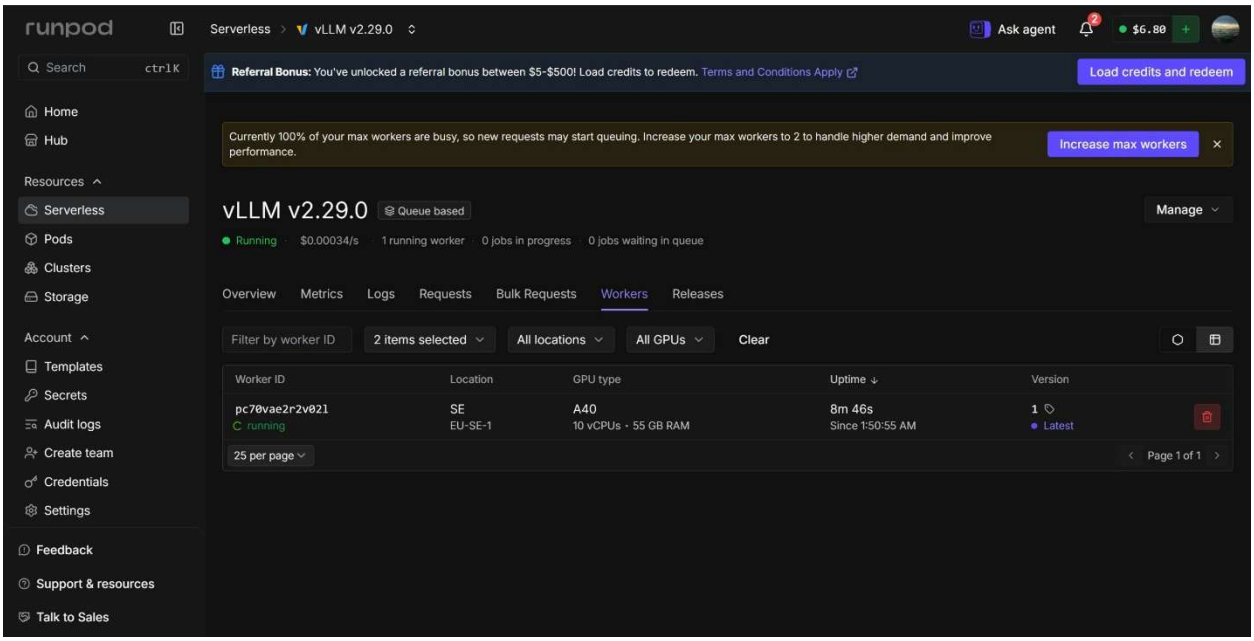
List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

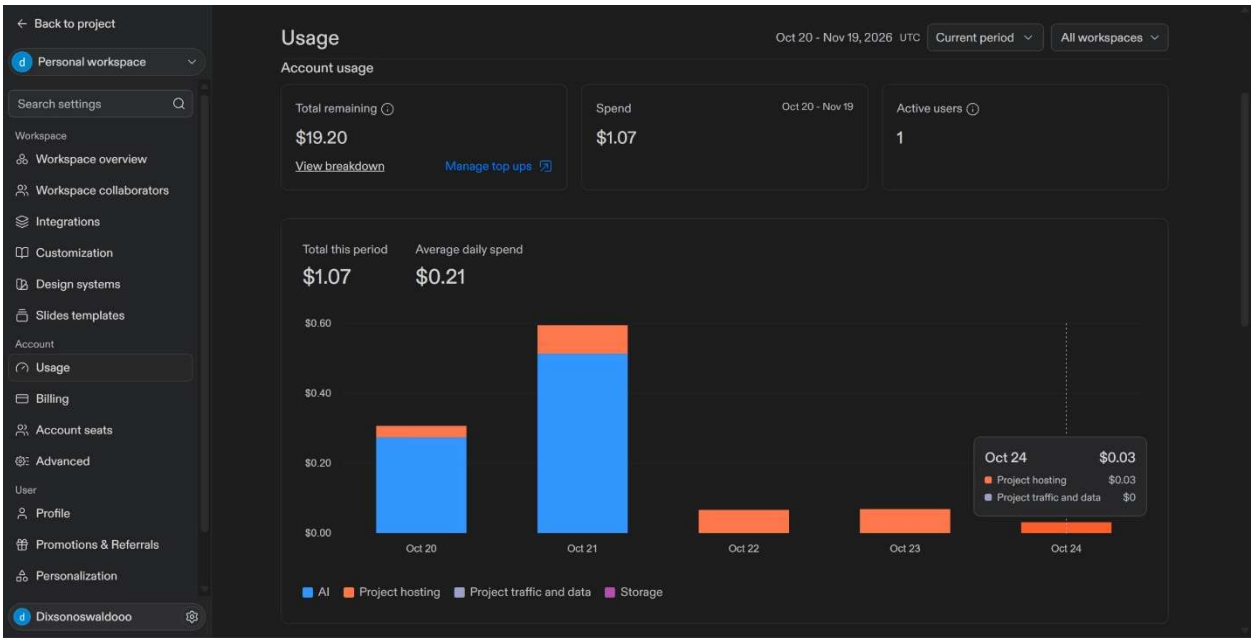
API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless (OpenAI-compatible endpoint) serving Gemma SEA-LION-v3-9B-IT	Reads free text leave reasons and suggests paid or unpaid leave, with a one sentence rationale and a confidence score	Sponsor provided credits (SDC)
WhatsApp Business Platform Meta	Not called in the prototype: messages are simulated in an in app outbox. Planned for production	Not used (simulated)
Orbyte attendance / e-leave	Not called in the prototype: replaced by synthetic attendance data	Not used (simulated)
Payment / banking API	None: “Pay all” records a simulated payment and reference number only	Not used

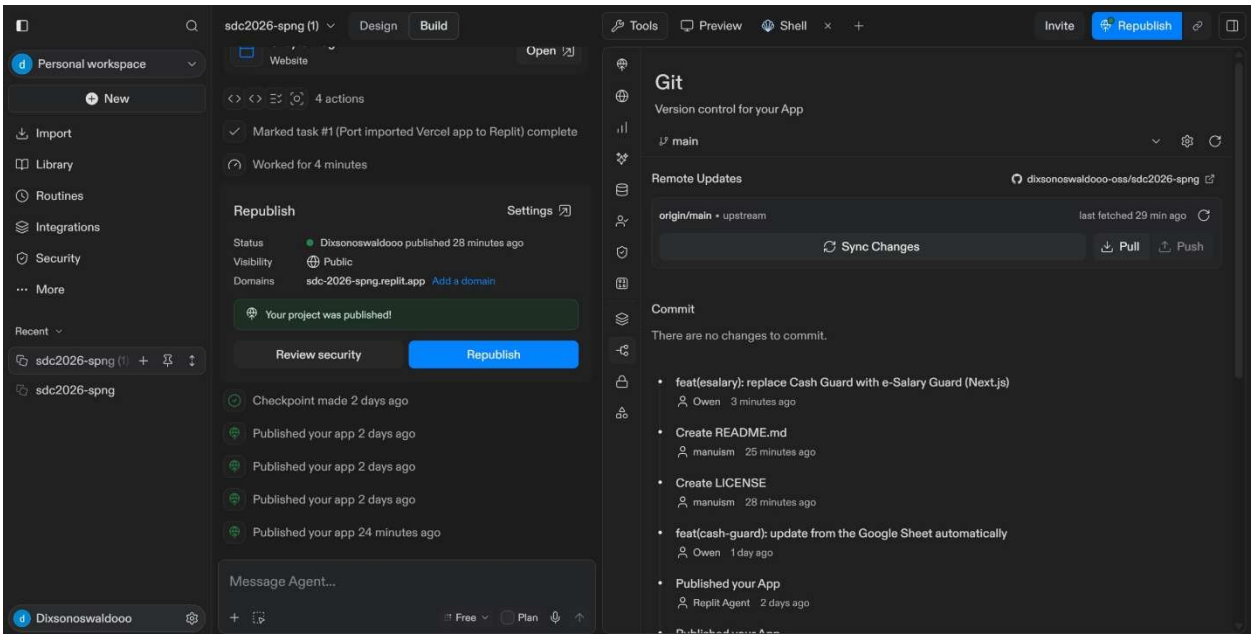
3.2 Cloud Platforms & Deployment

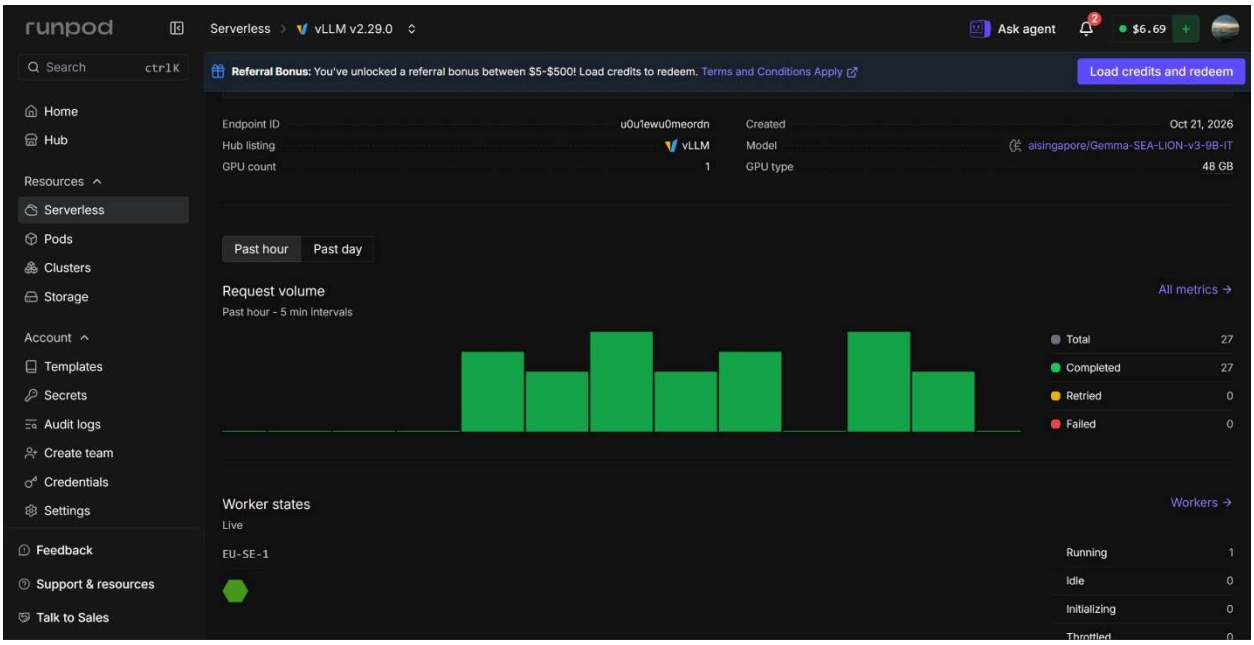
List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

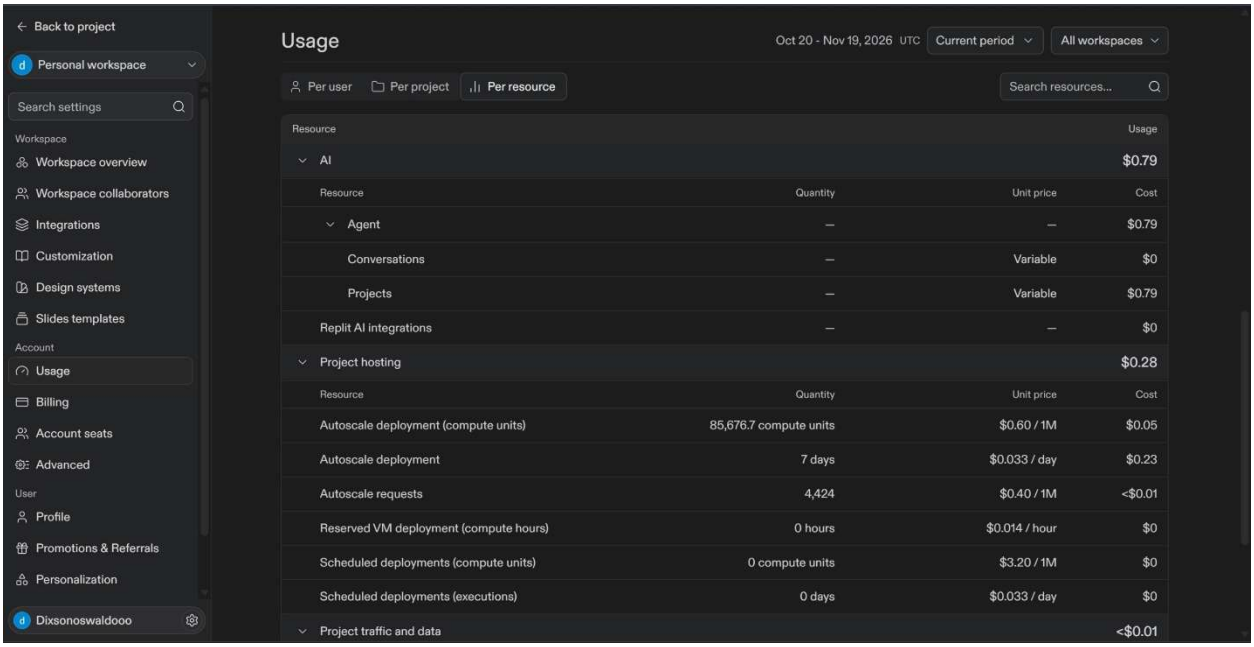
Service & Provider	Purpose in Project	Plan / Region
Replit Deployments	Hosts the Next.js app at sdc-2026-spng.replit.app	Autoscale deployment / Team Replit account / default region
Replit PostgreSQL	Managed database for the demo state (one row per demo space)	Included / default region
Replit Secrets	Stores RUNPOD_API_KEY, RUNPOD_ENDPOINT_ID, RUNPOD_MODEL and DATABASE_URL server side	Included
RunPod Serverless	GPU inference for SEA-LION	SDC sponsor credits
GitHub	Source control and deployment source	Free

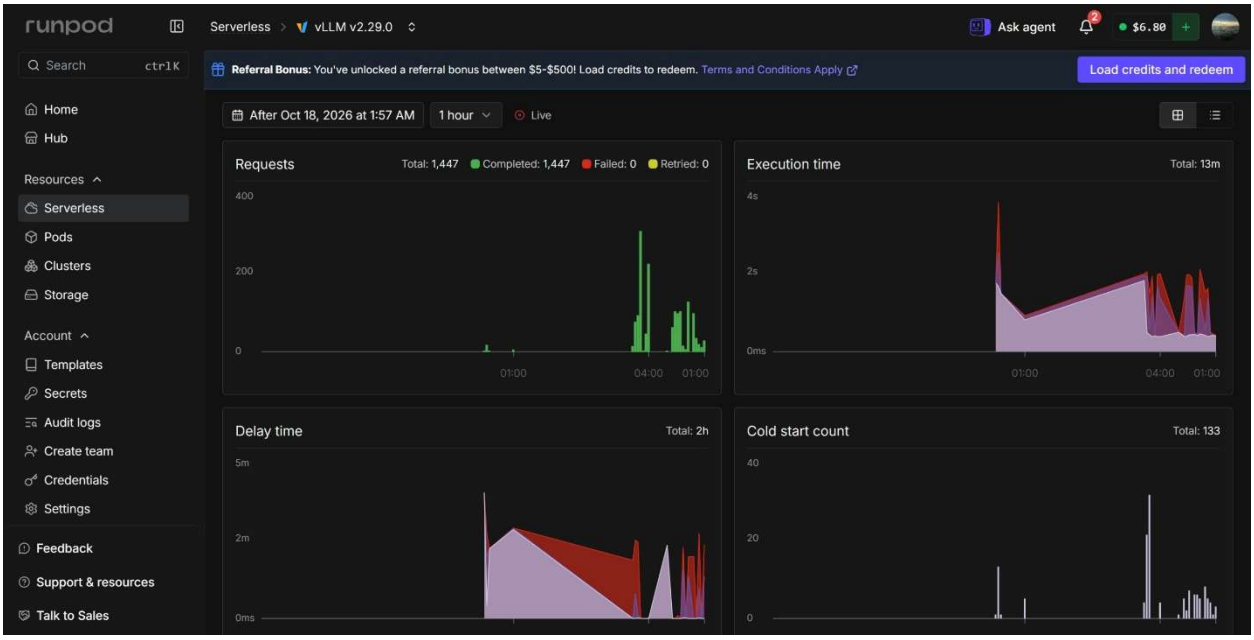

runpod
Serverless > vLLM v2.29.0
Ask agent
$6.80
Search ctrlK
Referral Bonus: You've unlocked a referral bonus between $5-$500! Load credits to redeem. Terms and Conditions Apply
Load credits and redeem
Home
Hub
Resources
Serverless
Pods
Clusters
Storage
Account
Templates
Secrets
Audit logs
Create team
Credentials
Settings
Feedback
Support & resources
Talk to Sales
Currently 100% of your max workers are busy, so new requests may start queuing. Increase your max workers to 2 to handle higher demand and improve performance.
Increase max workers
vLLM v2.29.0
Queue based
Manage
Running $0.00034/s 1 running worker 0 jobs in progress 0 jobs waiting in queue
Overview Metrics Logs Requests Bulk Requests Workers Releases
Filter by worker ID
2 items selected
All locations
All GPUs
Clear
Worker ID Location GPU type Uptime Version
pc70vae2r2v021
running
SE
EU-SE-1
A40
10 vCPUs · 55 GB RAM
8m 46s
Since 1:50:55 AM
1
Latest
25 per page
Page 1 of 1


Back to project
Personal workspace
Search settings
Workspace
Workspace overview
Workspace collaborators
Integrations
Customization
Design systems
Slides templates
Account
Usage
Billing
Account seats
Advanced
User
Profile
Promotions & Referrals
Personalization
Dixsonoswaldooo
Usage
Oct 20 - Nov 19, 2026 UTC
Current period
All workspaces
Account usage
Total remaining
$19.20
View breakdown
Manage top ups
Spend
Oct 20 - Nov 19
$1.07
Active users
1
Total this period
$1.07
Average daily spend
$0.21
$0.60
$0.40
$0.20
$0.00
Oct 20
Oct 21
Oct 22
Oct 23
Oct 24
Oct 24 $0.03
Project hosting $0.03
Project traffic and data $0
AI
Project hosting
Project traffic and data
Storage


sdc2026-spng (1)
Design
Build
Tools
Preview
Shell
Invite
Republish
Personal workspace
New
Import
Library
Routines
Integrations
Security
More
Recent
sdc2026-spng (1)
sdc2026-spng
Website
Open
4 actions
Marked task #1 (Port imported Vercel app to Replit) complete
Worked for 4 minutes
Republish
Settings
Status
Dixsonoswaldooo published 28 minutes ago
Visibility
Public
Domains
sdc-2026-spng.replit.app
Add a domain
Your project was published!
Review security
Republish
Checkpoint made 2 days ago
Published your app 2 days ago
Published your app 2 days ago
Published your app 2 days ago
Published your app 24 minutes ago
Message Agent...
Free
Plan
Dixsonoswaldooo
Git
Version control for your App
main
Remote Updates
dixsonoswaldooo-oss/sdc2026-spng
origin/main • upstream
last fetched 29 min ago
Sync Changes
Pull
Push
Commit
There are no changes to commit.
feat(esalary): replace Cash Guard with e-Salary Guard (Next.js)
Owen 3 minutes ago
Create README.md
manuism 25 minutes ago
Create LICENSE
manuism 28 minutes ago
feat(cash-guard): update from the Google Sheet automatically
Owen 1 day ago
Published your App
Replit Agent 2 days ago


runpod
Search
ctrlK
Home
Hub
Resources
Serverless
Pods
Clusters
Storage
Account
Templates
Secrets
Audit logs
Create team
Credentials
Settings
Feedback
Support & resources
Talk to Sales
Serverless
vLLM v2.29.0
Ask agent
$6.69
Referral Bonus: You've unlocked a referral bonus between $5-$500! Load credits to redeem. Terms and Conditions Apply
Load credits and redeem
Endpoint ID
u0u1ewu0meordn
Created
Oct 21, 2026
Hub listing
vLLM
Model
aisingapore/Gemma-SEA-LION-v3-9B-IT
GPU count
1
GPU type
48 GB
Past hour
Past day
Request volume
Past hour - 5 min intervals
All metrics
Total 27
Completed 27
Retried 0
Failed 0
Worker states
Live
EU-SE-1
Workers
Running 1
Idle 0
Initializing 0


Back to project
Personal workspace
Search settings
Workspace
Workspace overview
Workspace collaborators
Integrations
Customization
Design systems
Slides templates
Account
Usage
Billing
Account seats
Advanced
User
Profile
Promotions & Referrals
Personalization
Dixsonoswaldooo
Usage
Oct 20 - Nov 19, 2026 UTC
Current period
All workspaces
Per user
Per project
Per resource
Search resources...
Resource
Usage
AI
$0.79
Resource
Quantity
Unit price
Cost
Agent
–
–
$0.79
Conversations
–
Variable
$0
Projects
–
Variable
$0.79
Replit AI integrations
–
–
$0
Project hosting
$0.28
Resource
Quantity
Unit price
Cost
Autoscale deployment (compute units)
85,676.7 compute units
$0.60 / 1M
$0.05
Autoscale deployment
7 days
$0.033 / day
$0.23
Autoscale requests
4,424
$0.40 / 1M
<$0.01
Reserved VM deployment (compute hours)
0 hours
$0.014 / hour
$0
Scheduled deployments (compute units)
0 compute units
$3.20 / 1M
$0
Scheduled deployments (executions)
0 days
$0.033 / day
$0
Project traffic and data
<$0.01


runpod
Search ctrlK
Home
Hub
Resources
Serverless
Pods
Clusters
Storage
Account
Templates
Secrets
Audit logs
Create team
Credentials
Settings
Feedback
Support & resources
Talk to Sales
Serverless
vLLM v2.29.0
Ask agent
$6.80
Referral Bonus: You've unlocked a referral bonus between $5-$500! Load credits to redeem. Terms and Conditions Apply
Load credits and redeem
After Oct 18, 2026 at 1:57 AM
1 hour
Live
Requests
Total: 1,447
Completed: 1,447
Failed: 0
Retried: 0
400
200
0
01:00
04:00
01:00
Execution time
Total: 13m
4s
2s
0ms
01:00
04:00
01:00
Delay time
Total: 2h
5m
2m
0ms
Cold start count
Total: 133
40
20
0

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine tuned / self trained)	License / Terms
aisingapore/Gemma-SEA-LION-v3-9B-IT	AI Singapore (Hugging Face), served on RunPod Serverless with SDC sponsor credits	Pre-trained, instruction tuned; prompted to return a structured label (no fine tuning)	Gemma Terms of Use (built on google/gemma-2-9b)

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self collected / synthetic)	License	Personal Data? (Y/N)
Synthetic employee & pay settings (6 employees)	Generated by the team	Synthetic	Team owned	N
Synthetic attendance, Sep–Oct 2026 (Orbyte / e-leave simulation)	Generated by the team	Synthetic	Team owned	N
Leave-reason test cases	Written by the team	Synthetic	Team owned	N

No real or personal data is used. All names and WhatsApp numbers in the prototype are fictional.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

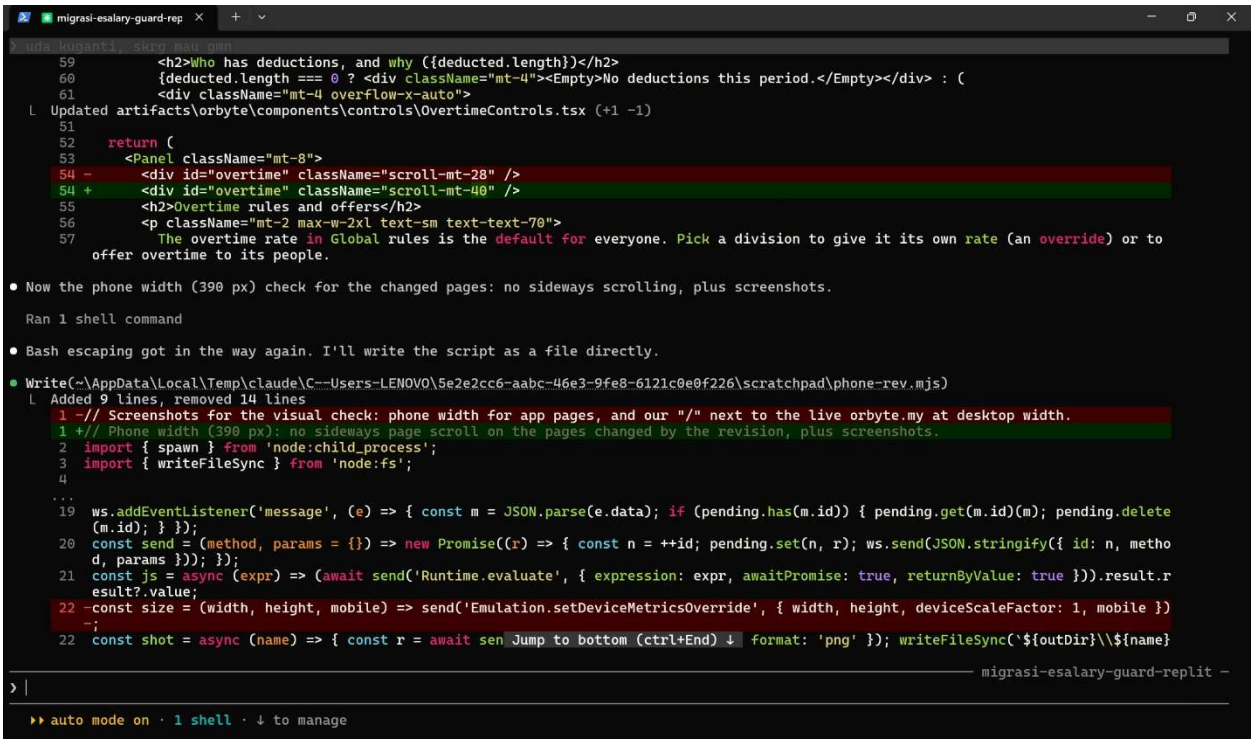
If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic) – Claude Opus 5.5	Writing and refactoring application code and unit tests	Major
Claude (claude.ai) – Claude Opus 5.5	Drafting the build specification and proposal text; research	Moderate
Replit Agent	Deployment and publishing	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI generated output:

[Maximum 100 words]

Every AI generated change is merged through GitHub only after the team member who owns that module has reviewed it; nobody merges code they cannot explain. The payroll engine is protected by unit tests whose expected values were calculated by hand, not taken from generated code. We run the full demo scenario end to end before each deployment, and every statistic and citation was checked against its original source. The team takes full responsibility for all submitted material.


migrasi-esalary-guard-rep
> uda kuganti, skrg mau gmn
59 <h2>Who has deductions, and why ({deducted.length})</h2>
60 {deducted.length === 0 ? <Empty>No deductions this period.</Empty> : (
61
└ Updated artifacts\orbyte\components\controls\OvertimeControls.tsx (+1 -1)
51
52 return (
53 <Panel className="mt-8">
54 -
54 +
55 <h2>Overtime rules and offers</h2>
56 <p className="mt-2 max-w-2xl text-sm text-text-70">
57 The overtime rate in Global rules is the default for everyone. Pick a division to give it its own rate (an override) or to offer overtime to its people.
● Now the phone width (390 px) check for the changed pages: no sideways scrolling, plus screenshots.
Ran 1 shell command
● Bash escaping got in the way again. I'll write the script as a file directly.
● Write(~\AppData\Local\Temp\claude\C--Users-LENOVO\5e2e2cc6-aabc-46e3-9fe8-6121c0e0f226\scratchpad\phone-rev.mjs)
└ Added 9 lines, removed 14 lines
1 -// Screenshots for the visual check: phone width for app pages, and our "/" next to the live orbyte.my at desktop width.
1 +// Phone width (390 px): no sideways page scroll on the pages changed by the revision, plus screenshots.
2 import { spawn } from 'node:child_process';
3 import { writeFileSync } from 'node:fs';
4
...
19 ws.addEventListener('message', (e) => { const m = JSON.parse(e.data); if (pending.has(m.id)) { pending.get(m.id)(m); pending.delete(m.id); } });
20 const send = (method, params = {}) => new Promise((r) => { const n = ++id; pending.set(n, r); ws.send(JSON.stringify({ id: n, method, params })); });
21 const js = async (expr) => (await send('Runtime.evaluate', { expression: expr, awaitPromise: true, returnByValue: true })).result.result?.value;
22 -const size = (width, height, mobile) => send('Emulation.setDeviceMetricsOverride', { width, height, deviceScaleFactor: 1, mobile });
22 const shot = async (name) => { const r = await sen Jump to bottom (ctrl+End) ↓ format: 'png' }); writeFileSync(`${outDir}\\${name}
migrasi-esalary-guard-replit
›
▸▸ auto mode on · 1 shell · ↓ to manage

5. Sponsor Provided Resources

Declare any official SDC sponsor provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod (Official SDC Partner / Sponsor)	Cloud credits & GPU compute instance	Used for hosting and performing inference for the SEA-LION (Southeast Asian Languages in One Network) AI language model, which automatically handles the classification of employee leave or time-off reasons on the backend.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Yes, resources from our sponsors are essential to the core functionality of the prototype. The credits and GPU infrastructure provided by RunPod allow the team to run the SEA-LION language model independently and securely on cloud servers, without relying entirely on paid, third-party commercial APIs. Nevertheless, our system architecture is designed to be modular utilizing server route handlers so that, if necessary, the model instances on RunPod can be replaced or integrated with alternative computing service providers without disrupting the primary logic of the payroll engine or the PostgreSQL database.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype uses only synthetic data: fictional employees, dummy WhatsApp numbers and generated attendance. **No real personal data is collected.** In a real deployment, the app would store names, job titles, salary settings, attendance times, leave reasons and phone numbers. Only the leave-reason text is sent to a third party (the SEA-LION model on RunPod); names, salaries and phone numbers are never included. Data sits in a managed PostgreSQL database reachable only through server side routes, role checks prevent employees from viewing others' records, and secrets are held in Replit Secrets. Demo data can be reset at any time and will be deleted after the hackathon. A production version would add full authentication, encryption at rest and a retention policy aligned with Malaysia's Personal Data Protection Act 2010.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

- **Misclassification or bias:** similar reasons may be labelled differently. *Mitigation:* HR approves every label, confidence below 0.6 is flagged, and overrides are logged.
- **Prompt injection:** reasons may contain instructions. *Mitigation:* text is treated as data, output is schema validated, and HR review is final.
- **Single provider dependency and rate limits.** *Mitigation:* a keyword fallback, labelled "Fallback".
- **Legal fit:** deduction rules vary by contract and must respect the Employment Act 1955 (s.24). Mitigation: rules are set by the employer, any deduction type can be switched off by setting its rate to zero, HR decides every leave label, and tax is out of scope.
- **Simulated integrations:** attendance, WhatsApp and payment are simulated.
- **Cold starts:** the serverless model can take 1–2 minutes to wake. Mitigation: requests are saved instantly with the keyword label, which the AI label replaces when it arrives.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Dixson Oswaldo Tanjaya, Team Leader, on behalf of SPNG
Date: 23 October 2026

Appendix A – System Block Diagram

Figure A1. End to end architecture of the e-Salary Guard prototype: role separated portals, server side payroll engine and AI classifier, managed PostgreSQL database, and simulated external integrations.

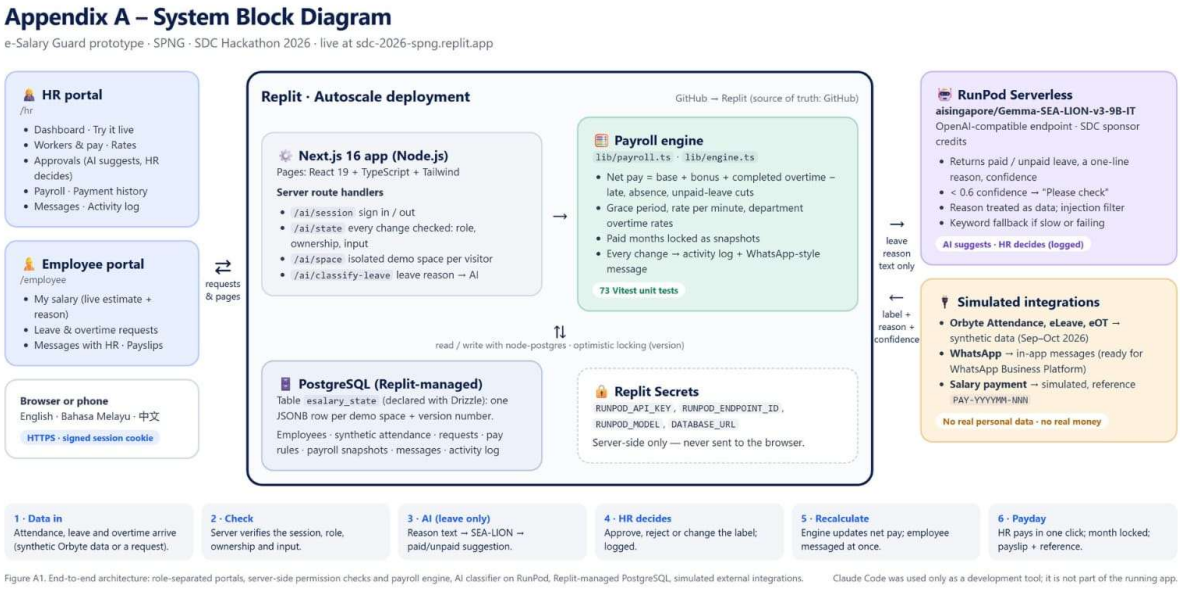


Figure A1. End-to-end architecture: role-separated portals, server-side permission checks and payroll engine, AI classifier on RunPod, Replit-managed PostgreSQL, simulated external integrations. Claude Code was used only as a development tool; it is not part of the running app.